

A per-basin Kalman filter as the benchmark for data-driven forecasts of GRACE terrestrial water storage

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Abstract. Data-driven forecasts of terrestrial water storage anomalies from GRACE and GRACE-FO are usually evaluated against climatology or persistence of the last observation. Neither accounts for the observation noise in each monthly basin value, which persistence carries unchanged into every lead time. We propose a per-basin Kalman filter with a first-order autoregressive state and additive observation noise as the benchmark for forecasts of deseasonalized basin storage. The filter has three parameters per basin, separates signal from noise before extrapolating, and remains defined through months without observations. On 234 global basins and five test folds issued between 2019 and 2026 on the CSR mascon solution, it reduces mean squared error relative to the stronger of two damped-persistence variants by 5.0 % at a lead time of 1 month and 8.8 % at 2 months. Refitting the filter with the observation-noise variance fixed at zero removes almost all of that margin. Scored against the filter in standardized units, in which every basin counts equally, the best model that uses only the basin’s own storage record and forcing is a linear ridge regression on a 12-month window of filtered state and ERA5-Land forcing. It gains a further 7.6 % at 1 month, none of the nonlinear models trained on the same inputs matched it, and its gain is confined to low-variance basins that pooled centimeter error barely weights. Against GRACE-FCast, a published deep-learning forecast that uses no lagged storage, the filter has lower error at 1 month on both CSR and JPL mascon solutions and higher error from 3 to 5 months onward, where the forcing history GRACE-FCast reads carries information the storage record does not. Damped persistence shares the 1-month win. Skill reported over persistence on this target should be re-evaluated against the filter.

1 Introduction

The GRACE and GRACE-FO missions have measured monthly changes in terrestrial water storage (TWS) since 2002, integrating groundwater, soil moisture, surface water, snow and ice into one observable that no in situ network can replicate (Tapley et al., 2004; Landerer et al., 2020). Anomalies of this quantity (TWSA) have a memory of several months, which makes them both a forecast target and a source of predictability. TWSA forecasts support drought early warning and water management (Li et al., 2026; Arsenault et al., 2020), flood-potential assessment (Reager et al., 2014), and the bridging of the two- to three-month latency of GRACE-FO data releases (Mo et al., 2025).

A growing number of studies forecast TWSA directly from data. Neural networks have been trained on GRACE and hydrometeorological series at basin scale (Ahmed et al., 2019), at grid scale globally (Li et al., 2020, 2024; Li and Kusche, 2026), for Europe with subsequent assimilation (Li et al., 2025), and for latency filling (Mo et al., 2025). Statistical time-series models

have been applied per basin (Ahi and Cekim, 2021), benchmark predictability has been quantified at decadal scale (Zhu et al., 2019), and physically based systems forecast TWS with seasonal meteorological forcing (Li et al., 2026; Getirana et al., 2020). Few of these studies compare their forecasts with any benchmark forecast. Skill is usually reported as correlation or Nash–Sutcliffe efficiency against the observations, whose implicit reference is the observed mean, or against another forecast system, as Li and Kusche (2026) do against SEAS5. None of the studies we examined scores against damped persistence. The one study that examines memory directly, the FEWS NET land data assimilation hindcast of Li et al. (2026), concludes that initial conditions dominate sub-seasonal to seasonal TWS skill, but measures that dominance with a lag-autocorrelation diagnostic rather than with a benchmark forecast.

In this paper we argue that persistence is the wrong benchmark for this target and propose a replacement. Basin-average TWSA is a state variable observed through a noisy instrument, and the observation errors are large for small and arid basins (Save et al., 2016; Landerer et al., 2020). A persistence forecast, damped or not, starts from the last observed month and carries the observation error of that month into every lead time. The benchmark we propose is a per-basin Kalman filter (Kalman, 1960) with a first-order autoregressive (AR(1)) state and additive observation noise. It separates signal from noise before extrapolating, has three parameters per basin, and needs no data beyond the storage series itself. Because it propagates its state through months without an observation, it also remains defined across the eleven-month gap between GRACE and GRACE-FO, where persistence has no last observation to start from.

The same argument has been made in sea-ice forecasting, where Niraula and Goessling (2021) showed that damped-persistence references that “do not exploit observations optimally” overstate the skill of dynamical forecast systems, and built a stronger reference from spatial damping. We make the analogous move for satellite-observed TWSA through signal and noise separation. Kalman methods are well established in the GRACE literature but serve other purposes: ensemble Kalman filters assimilate TWSA into land-surface models (Springer et al., 2026), state-space smoothers clean the gravity-field record (Kankanige et al., 2026), and integrated postprocessing separates the non-seasonal signal from noise in the monthly solutions (Zhang et al., 2025). None of these uses a univariate filter on the finished basin series as a benchmark forecast. The precedent for proposing a benchmark for TWS prediction is Zhu et al. (2019), who benchmarked decadal TWS predictability with an elasticity framework; we do the same at monthly lead times. Most directly, Nie et al. (2025) showed on 515 basins that deep architectures do not beat linear regression for TWS prediction when autoregressive information is excluded from the feature set by design. Their design has no lagged storage, so it has neither a persistence nor a filter benchmark. We show the complementary result: when lagged storage is included, how it is filtered decides the model comparison.

The paper addresses three questions. First, does the Kalman filter beat persistence and the conventional statistical benchmarks on deseasonalized basin TWSA, and if so, why? Second, scored against the filter, which models that use only a basin’s own storage record and forcing add skill, and does model architecture matter once the storage record is filtered? Third, how does the filter compare with a published deep-learning forecast, GRACE-FCast (Li and Kusche, 2026), on two independent mascon solutions? Section 2 describes the data, Sect. 3 the target, the evaluation design, the benchmarks, the models and the statistics, Sect. 4 the results, Sect. 5 their implications and limitations, and Sect. 6 concludes.

2.1 GRACE and GRACE-FO mascon solutions

The primary storage data set is the CSR RL0603M mascon solution (Save et al., 2016) on a 0.25° grid, April 2002 through May 2026, in centimeters of equivalent water height; the product version is RL06.2. Each monthly solution is assigned to its official calendar month using the product’s documented list of missing months rather than by the midpoint of its data span, which would mislabel two solutions. The second storage data set is the JPL RL06.3Mv04 mascon solution with the coastline resolution improvement filter applied, sampled at 0.5° , April 2002 through June 2026, with the product’s scale factors applied and its official missing-month metadata used. Both products have a native resolution of roughly 3° equal-area mascons and both are released with correlated-error filtering already applied. They are independent processings of the same satellite observations, so a property of the benchmark that depended on the processing rather than on the observations would not be expected to survive the change of product.

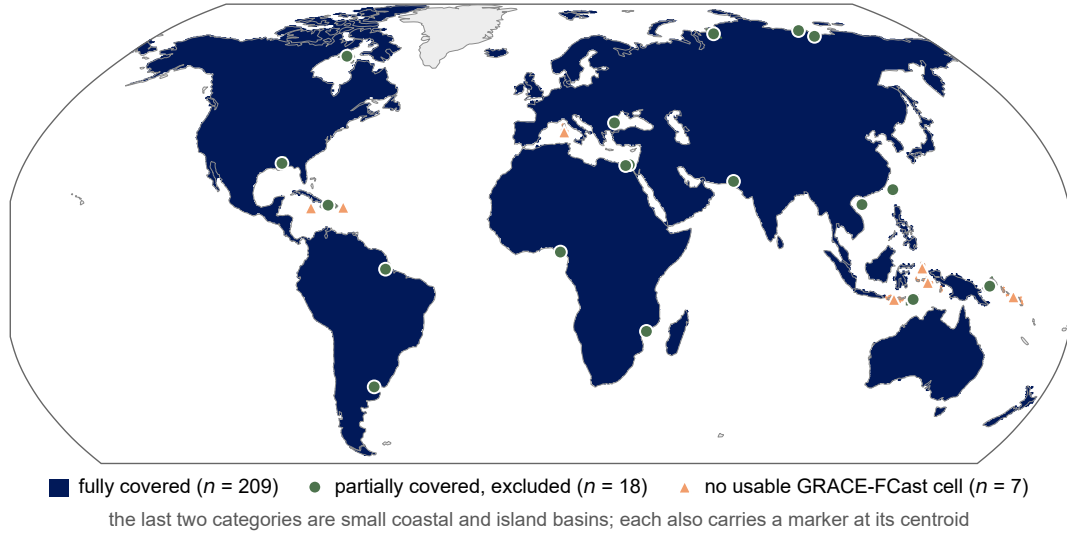
2.2 Basins

Basins follow a HydroSHEDS-based Level-3 basin and mascon partition (Lehner and Grill, 2013). Each basin series is the cosine-latitude area-weighted mean of the grid cells in its mask. Of the 284 basins in the partition we exclude 46 ice-sheet basins in Antarctica and Greenland and 4 water bodies, leaving 234 basins on the CSR product. On the JPL product six small-island masks contain no valid scaled cells, so 228 basins are kept there. Figure 1 maps the basins and marks the subsets used in the comparison with GRACE-FCast (Sect. 3.6).

2.3 ERA5-Land forcing and climate indices

We use 11 ERA5-Land variables (Muñoz-Sabater et al., 2021) at 0.5° , regridded server-side from 0.1° , for January 2001 through May 2026: total precipitation, evaporation, total, surface and subsurface runoff, volumetric soil water in four layers, 2-m temperature, and snow depth water equivalent. Accumulated fluxes are converted from mean daily rates to monthly totals, evaporation is sign-flipped so that positive values mean moisture loss, temperature is converted to degrees Celsius, and snow depth to centimeters of water equivalent. Basin aggregation uses the same cosine-latitude weighted means as the storage aggregation, mapping each 0.25° mask cell to its nearest 0.5° ERA5 cell. Because ERA5-Land is defined over land only, coastal and island basins aggregate their forcing from the land fraction of their mask. Of the 234 basins, 79 draw less than 90 % of their mask weight from valid ERA5 land cells and 13 less than 50 % (minimum 24.5 %); no basin is excluded on this ground. Twelve monthly climate indices from NOAA PSL enter one benchmark model only (Sect. 3.3); two of them, AMM and PDO, end before the test window and are dropped from it.

(a) CSR mascons: the 234-basin sample and the fully covered comparison set



(b) JPL mascons: the fully covered comparison set

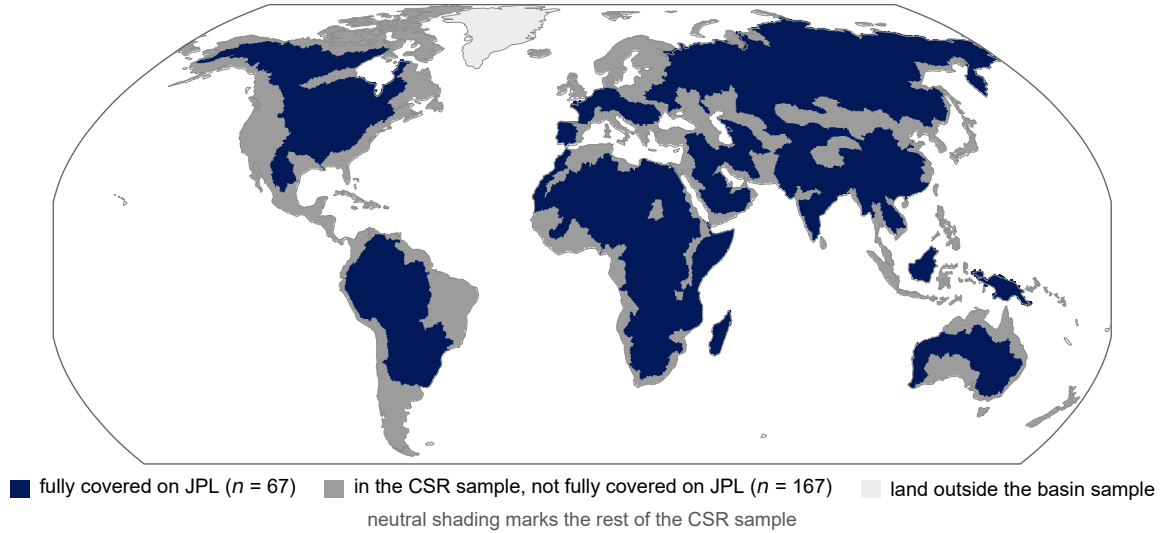


Figure 1. The basin sample. (a) The 234 basins kept on the CSR product, colored by their role in the comparison with GRACE-FCast (Sect. 3.6): 209 basins are fully covered by both the mascon grid and the forecast grid and enter the comparison, 18 have partial coverage and are excluded by the coverage rule, and 7 have no usable GRACE-FCast cell. (b) The JPL product: the 67 basins that pass the coverage rule there. The remaining basins of the CSR sample are shaded neutral; the three-way split of (a) was not evaluated on JPL. Land outside the basin sample is pale grey in both panels.

2.4 GRACE-FCast

For the external comparison we use the public GRACE-FCast hindcast of Li and Kusche (2026), downloaded from PANGAEA (Li, 2025). It is released as three members built on the JPL, CSR and GSFC mascons. We use the CSR member against our CSR target and the JPL member against our JPL target, so that each comparison shares a gravity solution. The CSR member holds monthly 1° global forecasts at lead times of 1 to 12 months, initialized December 2009 through May 2024 (174 initializations); the JPL member holds 173 initializations through April 2024. GRACE-FCast is a long short-term memory network that forecasts the interannual and sub-seasonal components of TWSA from antecedent hydrometeorological and oceanographic predictors, with correlation-based spatial shifting of the predictors. By design it takes no lagged storage as input, and its seasonal and trend components are extrapolated from the previous year and added back (Li and Kusche, 2026). We evaluate both the full product and its non-seasonal component (Sect. 3.6).

The skill reported by Li and Kusche (2026) is for the average of the three members, and their training-residual diagnostic ranks the CSR member as the least certain of the three (3.0 against 2.8 cm), so our results characterize the members we score, not their headline product. Their JPL member was trained against JPL RL06.1, whereas our JPL target is RL06.3Mv04, so that comparison crosses a release. Their training also fills the 2017–2018 mission gap with the reconstruction of Yin et al. (2023), which our models never see.

3 Methods

3.1 Forecast target

For each basin and each evaluation fold we fit, on the training window only, a climatology consisting of a linear trend plus annual and semiannual harmonics (six coefficients, ordinary least squares). The frozen coefficients are used to deseasonalize both the training and the test data, and the residual is divided by its training-window standard deviation. The forecast target is this deseasonalized, per-basin standardized residual. It holds the interannual anomalies that drought and flood applications need, and standardizing it makes every basin count equally in pooled metrics. Section 4.5 and Appendix C report the same forecasts in centimeters, where large basins dominate.

3.2 Evaluation design

We use five expanding-window folds. Their issue windows begin in June 2019, November 2020, April 2022, September 2023 and February 2025, and together cover June 2019 through April 2026 at a 1-month lead, with targets reaching May 2026. Fold membership is defined by the forecast issue date. Within each fold, every climatology, filter parameter, model coefficient and standardization constant is fit only on observations before the fold’s first issue date, and every test forecast is issued at or after that date. A forecast issued before the fold start but targeting a month after it belongs to neither set. Defining membership by target date instead would let a lead- h forecast use transforms fit up to $h - 1$ months after its issue date. Earlier test windows are not usable, because the 2017–2018 gap destroys the lag windows needed to forecast across it. Every model is scored on the

same rows at a given lead time, so the sample on the CSR product is $n = 19,422, 19,188, 18,954, 18,720, 18,486$ and $18,252$ basin-months at lead times of 1 to 6 months. On the JPL product, whose record runs one month longer, the 228 basins carry 84 issue months at a 1-month lead against 83 on CSR, so $n = 19,152$ there. Issue time is defined relative to the last available observation, so the release latency of GRACE-FO is not modeled and all results are retrospective potential skill.

3.3 Benchmark forecasts

Every model is compared with an ordered set of benchmark forecasts of increasing strength:

- climatology, the zero-anomaly forecast;
- persistence, the last observed anomaly carried forward unchanged;
- damped persistence in two variants: the ρ^h variant multiplies the last anomaly by a per-basin AR(1) coefficient ρ raised to the power of the lead time, and the regression variant regresses the lead- h target directly on the last anomaly;
- a pooled ridge regression on own-basin lagged anomalies (penalty 1; lags 0, 1, 2, 5 and 11 months; one model for all basins), and the same ridge with the lagged climate indices of Sect. 2.3 added;
- a per-basin ridge on the same lags, refit separately for each basin.

We always score against the stronger damped-persistence variant at each lead time: on CSR the ρ^h variant at 1 month and the regression variant at 2 to 6 months, on JPL the ρ^h variant at 1 and 2 months and the regression variant at 3 to 6 months (Appendix B). Scoring against the weaker variant, which trails by up to 15 %, would inflate every other model’s skill. Each table and figure states the variant used.

3.4 The Kalman filter benchmark

The proposed benchmark models each basin’s deseasonalized residual y_t as an AR(1) latent signal x_t observed with additive white noise:

$$x_t = \rho x_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, q), \quad (1)$$

$$y_t = x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, r). \quad (2)$$

The three parameters (ρ, q, r) are estimated per basin and fold by maximum likelihood on training months only, using L-BFGS-B from several starting points. The filter runs forward through the training and test months and updates its state estimate $x_{t|t}$ at each observed month with the Kalman gain, which depends on ρ and on the ratio q/r . At a month with no observation, including the eleven months of the mission gap, the filter skips the update and propagates its state and state uncertainty forward; nothing is interpolated or filled. The forecast for lead time h issued at month t is

$$\hat{y}_{t+h|t} = \rho^h x_{t|t}, \quad (3)$$

which is damped persistence applied to the filtered state instead of to the raw observation. Throughout the paper we refer to this forecast as the Kalman filter (KF). It is not a data-assimilation scheme: nothing is assimilated into any model, and the only state estimated is the basin’s own scalar signal.

Two variants of the filter serve to explain its margin over damped persistence. In the *noise-free variant* ($r = 0$) the identical model is refit by the identical likelihood with r fixed at zero. This forces the Kalman gain to one, so the state tracks the raw observation and the forecast reduces to damped persistence with a maximum-likelihood ρ ; the difference between the two filters is the contribution of the noise term alone. In the *two-variance variant* the observation equation has separate noise variances for months before January 2018 and for GRACE-FO months from then on, so the parameter vector is $(\rho, q, r_{\text{GRACE}}, r_{\text{FO}})$. A fold whose training window holds fewer than 12 GRACE-FO months cannot separate the second variance from q ; such fits fall back to the single-variance model.

The split between signal variance q and observation variance r is set by the likelihood and should not be read as a measurement of instrument noise. Fast hydrological variability that an AR(1) state cannot carry forward, and errors in the deseasonalization, are absorbed into r alongside instrument noise, and in a minority of fits the likelihood pushes r to its lower bound: 259 of the 1170 basin-fold fits do so. We therefore use “observation noise” in the model’s sense, as whatever the filter treats as unpredictable month-to-month variation, rather than as an estimate of instrument error.

3.5 Forecast models built on the filter

We call the following models own-basin models because each reads only its own basin’s storage record and forcing. Every one of them predicts the error of the Kalman filter forecast, and its own forecast is the filter forecast plus that predicted error. Table 1 lists the models and the abbreviations used for them throughout; in the abbreviations, R denotes a ridge correction, the number the months of filtered-state history it reads, and E the ERA5-Land forcing. Four models are linear. KF-R1 regresses the error on the current filtered state alone. KF-R12 regresses it on the last twelve months of the filtered state. KF-R12E adds the twelve months of all 11 ERA5-Land anomaly series, giving 12 by 12 features. KF-R1E instead adds the ERA5-Land anomalies at lags of 0, 1 and 2 months to the current filtered state. Each ERA5-Land basin series is deseasonalized with the same training window trend and harmonic fit as the storage series (Sect. 3.1), standardized by its training-window standard deviation and clipped at ± 10 standard deviations before it enters any model. All four ridges are pooled across basins with penalty 1 on train-standardized features. Three models are nonlinear and use the same information. The LSTM is a shared-encoder long short-term memory network (Hochreiter and Schmidhuber, 1997) over the same twelve-month window of filtered state and the 11 ERA5-Land channels. It is trained with a time-ordered split that holds out the last 15 % of training months for early stopping, and its two seeds are reported as a seed ensemble. The MLP is a multilayer perceptron (hidden layers of 64 and 32 units, early stopping) on the KF-R12E feature vector, with two seeds. The residual MLP is a two-stage model, KF-R1E followed by an MLP of the same size trained on the first stage’s training residual, with three seeds. Finally, two neighbor models, not adopted (Sect. 4.6), add to KF-R1 the filtered state of one other basin propagated to the target month, with that basin chosen by correlation or by distance. Five of these forecasts are also reported in centimeters (Sect. 4.5 and Appendix C): the stronger damped-persistence variant, KF, KF-R1, KF-R12 and KF-R12E.

Table 1. Forecasts evaluated in this paper, with the section that defines each. Appendix A gives the identifier of each in the code and data repository.

Name	Definition
Climatology	Zero-anomaly forecast (Sect. 3.3).
Persistence	Last observed anomaly carried forward (Sect. 3.3).
Damped persistence	Last anomaly damped toward zero; ρ^h and regression variants, always scored against the stronger at each lead time (Sect. 3.3).
Pooled lag ridge	Ridge regression on own-basin lagged anomalies, one model for all basins; the index variant adds lagged climate indices (Sect. 3.3).
Per-basin lag ridge	The pooled lag ridge’s features, refit separately for each basin (Sect. 3.3).
KF	The proposed benchmark: damped persistence of the filtered state from the per-basin AR(1) plus observation-noise model (Sect. 3.4).
KF, $r = 0$	The noise-free variant, refit with the observation variance fixed at zero (Sect. 3.4).
KF, two variances	The two-variance variant, with separate observation variances for GRACE and GRACE-FO months (Sect. 3.4).
KF-R1	KF plus a ridge correction on the current filtered state (Sect. 3.5).
KF-R12	KF plus a ridge correction on the 12-month history of the filtered state (Sect. 3.5).
KF-R12E	KF-R12 with the 12-month history of the 11 ERA5-Land anomalies added (Sect. 3.5).
KF-R1E	KF plus a ridge correction on the current filtered state and ERA5-Land anomalies at lags of 0 to 2 months (Sect. 3.5).
LSTM	Shared-encoder LSTM over the 12-month history of filtered state and ERA5-Land channels; two-seed ensemble (Sect. 3.5).
MLP	Multilayer perceptron on the KF-R12E features; two seeds (Sect. 3.5).
Residual MLP	KF-R1E followed by an MLP on its training residual; three seeds (Sect. 3.5).
Neighbor ridge	KF-R1 plus one other basin’s propagated filtered state, selected by correlation or by distance (Sect. 3.5; not adopted, Sect. 4.6).
GRACE-FCast, full and non-seasonal	The published forecast of Li and Kusche (2026), with and without its extrapolated seasonal and trend terms (Sects. 2.4 and 3.6).

3.6 Comparison with GRACE-FCast

The GRACE-FCast hindcast (Li, 2025) is aggregated to our basin geometry with exact 0.25° to 1° cell-containment weights. Its forecasts are moved into each fold’s deseasonalized, standardized space using our train-window climatology plus a per-fold, per-basin, per-lead mean offset estimated on pre-test data only. The full product is scored after our climatology is subtracted, so any difference between their seasonal and trend convention and ours is absorbed there. The non-seasonal component, which is their interannual plus sub-seasonal forecast, is scored in the same space; this removes the advantage that their monthly re-

extrapolation of the seasonal cycle and trend would otherwise carry. Every model is scored on the identical set of basin-months at every lead time. On CSR it spans 60 test months: folds 1 to 3 in full, 9 of the 17 issue months of fold 4, and none of fold 5, because GRACE-FCast’s final initialization, May 2024, falls inside fold 4. On JPL, whose final initialization is one month earlier, it spans 59 months.

We define two basin sets. The matched set keeps every basin with at least one usable GRACE-FCast cell, which is 227 of the 234 basins on CSR and 227 of 228 on JPL. The fully covered set applies one rule to both products: a basin enters only if it fully contains every 0.25° mask cell of at least one native mascon of the product being scored, and all sixteen cells of at least one finite 1° GRACE-FCast cell. This is a literal containment test rather than an area or coverage proxy. On CSR, 229 basins contain a full native mascon, 211 contain a full forecast cell, and 209 satisfy both; on JPL, whose native mascons are coarser on the sampled grid, 67 do. We report the fully covered set. On the matched set, the standardized 1-month error of the full product on JPL is 5.47 against 0.93 on the fully covered set, because our standardization uses the residual standard deviation of our own product, which is small in basins that the coarser forecast grid only partly covers. Matched-set values therefore measure grid mismatch, not forecast accuracy.

3.7 Verification statistics

The primary metric is the pooled root-mean-square error (RMSE) in standardized units. We report it as the mean-squared-error skill score of a forecast A against a stated benchmark B ,

$$S_{A|B} = 1 - \frac{\text{MSE}_A}{\text{MSE}_B}, \quad (4)$$

in percent, so that a positive value means A has the lower error. Every skill in this paper is stated with our forecast first, so a positive value always means our forecast is better. Pooled comparisons use the Diebold–Mariano test (Diebold and Mariano, 1995) on the monthly cross-basin mean loss differential, with the small-sample correction of Harvey et al. (1997) and a Newey–West variance at maximum lag $\max(h-1, 1)$ (Newey and West, 1987). Confidence intervals use a moving-block bootstrap (Künsch, 1989) of the monthly losses with block length $\max(3, h)$ and 2000 draws, so that the block covers the lag- $(h-1)$ dependence of overlapping h -step forecasts. Per-basin significance uses per-basin Diebold–Mariano statistics on the scored test months with Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ (Benjamini and Hochberg, 1995), following the field-significance reasoning of Wilks (2016); basins that are helped and basins that are hurt are always counted separately. In the text we state whether a difference is significant at $p < 0.05$ and quote the p value only where a result is marginal; exact p values are given in the tables.

4 Results

4.1 Benchmark forecasts

Table 2 and Fig. 2a compare the benchmark forecasts on the CSR product. KF has lower error than the stronger damped-persistence variant at every lead time, by 5.0 % at 1 month, 8.8 % at 2 months and 2.5 to 5.6 % at 3 to 6 months ($p < 10^{-6}$ at

Table 2. Benchmark forecasts on the CSR product: pooled skill (%) against the stronger damped-persistence variant at each lead time h (months), on the same basin-months for every forecast (234 basins, five folds; n as in Sect. 3.2). Positive is better. The reference is the ρ^h variant at $h = 1$ and the regression variant at $h = 2$ to 6. The best value in each column is in bold. Appendix B gives the standardized RMSE and the Diebold–Mariano p value for every row.

Forecast	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
Climatology	−125.6	−70.7	−58.9	−51.4	−42.7	−33.6
Persistence	−20.7	−22.7	−21.4	−19.5	−16.0	−15.8
Damped persistence, weaker variant	−1.3	−0.6	−6.4	−11.8	−12.0	−10.0
Damped persistence, stronger variant	0.0	0.0	0.0	0.0	0.0	0.0
Pooled lag ridge	+1.2	+4.8	+5.1	+3.6	+3.2	+2.5
Pooled lag ridge, index variant	+1.0	+4.5	+5.1	+3.7	+3.2	+2.4
Per-basin lag ridge	+3.1	+5.9	+4.3	+2.1	+0.4	−0.3
KF	+5.0	+8.8	+5.6	+3.1	+2.5	+3.6
KF-R1	+4.8	+7.9	+6.0	+5.4	+5.1	+5.2
KF-R12	+5.7	+10.1	+7.3	+4.8	+4.0	+4.2
KF-R12E	+12.2	+13.7	+9.6	+6.2	+4.6	+4.5

every lead time; Appendix B). The margin peaks at 2 months, and the 95 % bootstrap interval on the 1-month margin is +3.9 to +6.1 %. KF also beats the per-basin lag ridge, the strongest conventional statistical benchmark at short lead times, by 1.0 to 3.9 %; the difference is significant ($p < 0.02$) at five of the six lead times, and at 4 months it is marginal ($p = 0.078$), although the bootstrap interval excludes zero (+0.02 to +2.20 %). Against the pooled lag ridge KF is ahead by 3.9 and 4.2 % at 1 and 2 months ($p < 10^{-5}$) and indistinguishable from it at 3 to 6 months ($p \geq 0.18$). A pooled ridge with five lag coefficients shared across 234 basins extrapolates well at long lead times, but it does not filter the most recent observation, which is where the 1- and 2-month margins arise.

Figure 3 shows the 1-month forecasts in four basins. In the Saharan and Algerian Atlas and in the East Nile Delta, where the month-to-month scatter is large relative to the interannual signal, damped persistence follows each excursion of the last observation while the filter tracks the slower signal beneath it. In the Danube the two forecasts are close, and in the Amazon, where the signal is large and smooth, they are nearly indistinguishable and damped persistence is marginally better. Across the sample, KF has the lower 1-month RMSE in centimeters in 158 of the 234 basins and damped persistence in 76, with the largest gains in arid and small basins (Fig. 4c), where the formal error estimates of the mascon solutions are also largest (Save et al., 2016).

The gap between raw and damped persistence, 16 to 23 %, is the cost of not damping; filtering the state before damping it adds a further 5.0 to 8.8 % at 1 and 2 months. The statistical benchmarks without noise separation lie between damped persistence and the filter: the per-basin lag ridge, with five coefficients per basin, never reaches the three-parameter filter and

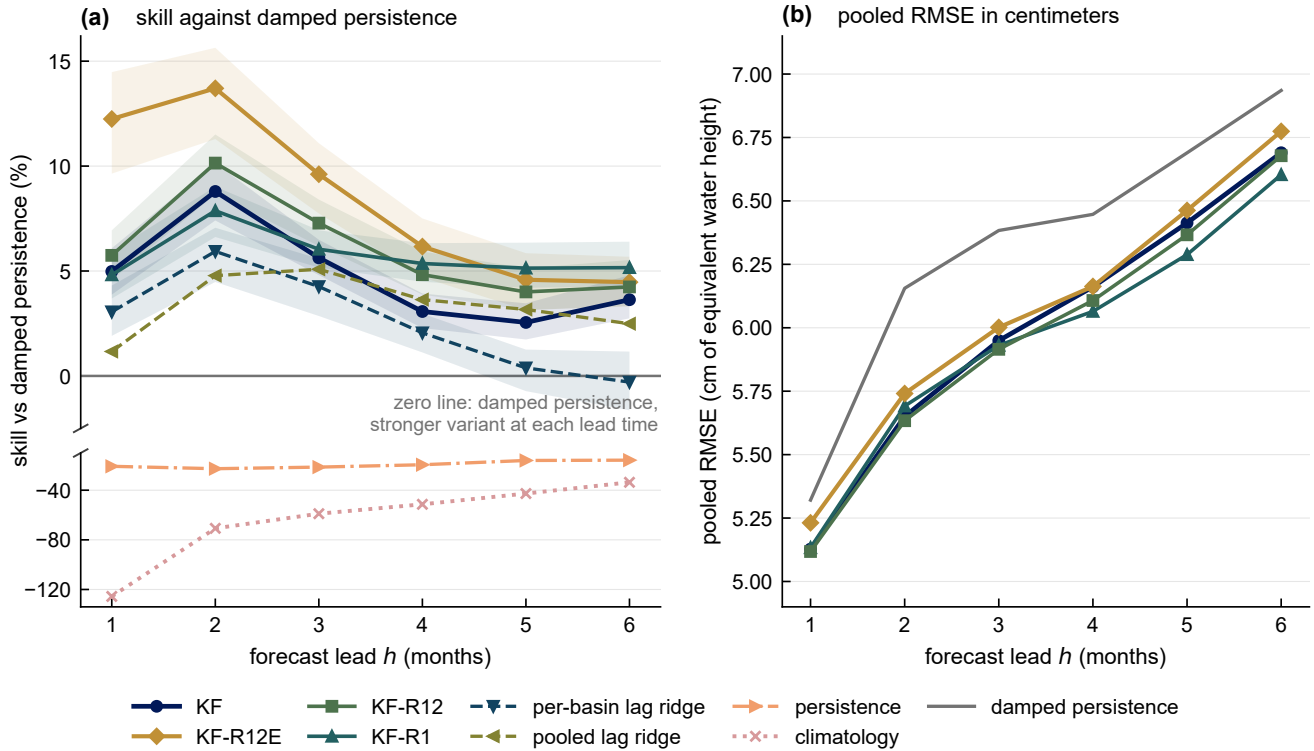


Figure 2. Benchmark forecasts on the CSR product. (a) Pooled skill (%) against the stronger damped-persistence variant (the ρ^h variant at a 1-month lead time and the regression variant at 2 to 6 months) for KF-R12E, KF, KF-R12, KF-R1, the per-basin lag ridge, the pooled lag ridge, persistence and climatology, the last two on a broken axis. Shading gives moving-block bootstrap 95 % confidence intervals for KF-R12E, KF, KF-R12, KF-R1 and the per-basin lag ridge. The gap between the zero line and KF is the gain from filtering the most recent observation before damping it. (b) Pooled RMSE in centimeters of equivalent water height by lead time for the five forecasts also reported in centimeters: damped persistence (stronger variant), KF, KF-R1, KF-R12 and KF-R12E. In this weighting the largest basins dominate and the ranking changes (Sect. 4.5).

235 by 6 months has fallen below damped persistence. The filter is the strongest forecast in the benchmark set and requires no data beyond the storage series.

On the JPL product, with 228 basins and the same folds, KF's margin over the stronger damped-persistence variant holds at 2 to 6 months (+1.5 to +6.3%, $p < 0.02$) but is a tie at 1 month (+0.6%, $p = 0.60$). Against the per-basin lag ridge the pattern is the same: KF is behind at 1 month (-0.66% , $p = 0.46$) and ahead by +2.11 to +6.67% at 2 to 6 months ($p < 0.03$;
 240 Table B2). The 1-month tie on JPL is consistent with the mechanism of Sect. 4.2: if JPL's coarser mascons and coastline filter leave less month-to-month noise in the basin series, the filter has less to remove at the lead time where noise matters most. The smaller standardized damped-persistence error on JPL (0.67 against 1.08 on CSR at a 1-month lead time) is consistent with that reading, which we do not test directly (Sect. 5.3).

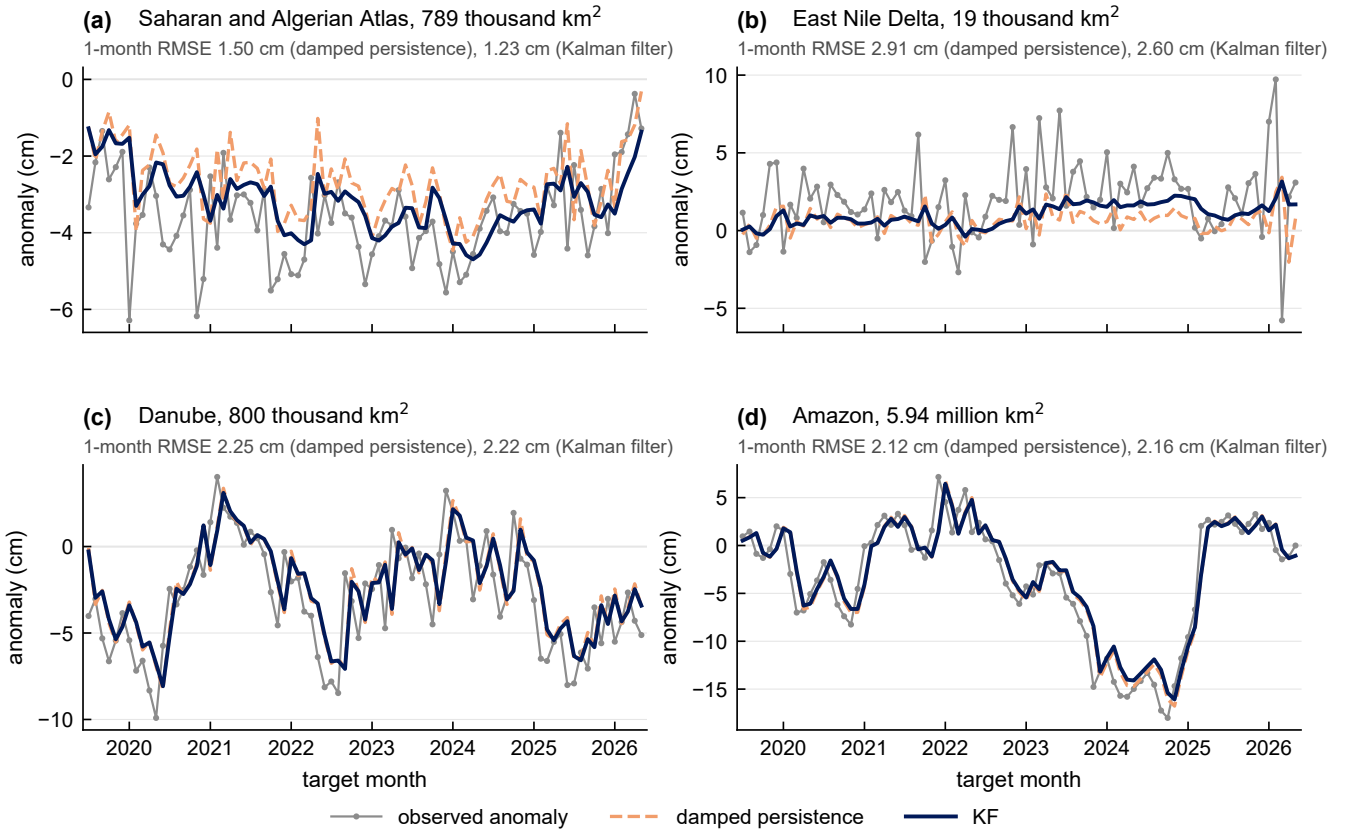


Figure 3. Observed deseasonalized storage anomaly (grey) and the 1-month forecasts of damped persistence (ρ^h variant, dashed) and KF (solid) in four basins, in centimeters of equivalent water height, over the full test window. The panels span the range of per-basin outcomes at a 1-month lead time: (a) and (b) two arid basins with noisy observations, where the filter gains most; (c) a mid-latitude river basin where the gain is a few percent; (d) the Amazon, where damped persistence is marginally better. Each panel states the 1-month RMSE of both forecasts over the window.

4.2 Why the filter beats damped persistence

Two ablations attribute the margin (Fig. 4). The noise-free variant, the same model refit with $r = 0$, loses to the full filter at every lead time, by 5.35% at 1 month, rising to 12.65% at 5 months ($p < 10^{-13}$ at every lead time). Scored against the regression variant at every lead time, the noise-free variant is +0.90% ahead at 1 month and 0.93 to 11.57% behind at 2 to 6 months; against the ρ^h variant, the stronger one at 1 month, it is 0.40% behind. At a 1-month lead time the AR(1) structure without the noise term is therefore within 1% of damped persistence in either direction, and the two mean-squared-error ratios, +5.35% for the noise term and -0.40% for the estimate of ρ , multiply to the +4.98% of Table 2: the margin is almost entirely

Table 3. Own-basin models against the Kalman filter: pooled skill (%) against KF at each lead time h (months) on the same basin-months (upper value), with the Diebold–Mariano p value below it. Positive means the model improves on the filter; the best value in each column is in bold.

Forecast	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
KF-R1	−0.17 1.8×10^{-6}	−1.01 1.9×10^{-11}	+0.44 1.1×10^{-14}	+2.36 1.6×10^{-13}	+2.66 5.9×10^{-14}	+1.59 2.3×10^{-17}
KF-R12	+0.81 0.0035	+1.48 4.9×10^{-6}	+1.76 2.9×10^{-10}	+1.81 7.5×10^{-10}	+1.49 3.4×10^{-7}	+0.63 0.015
KF-R12E	+7.65 1.1×10^{-8}	+5.39 1.1×10^{-8}	+4.22 3.0×10^{-8}	+3.19 5.2×10^{-6}	+2.09 9.0×10^{-4}	+0.87 0.12

The KF-R1 p values are two-sided tests of a difference that is negative at $h = 1$ and 2 and positive at $h = 3$ to 6.

the noise term. Because r absorbs whatever an AR(1) state cannot carry forward (Sect. 3.4), the ablation attributes the margin to filtering within the model class rather than to a particular physical noise source.

The second ablation asks whether the margin depends on treating GRACE and GRACE-FO as one instrument. The two-variance variant, with a separate observation variance for GRACE-FO months, is worse than the single-variance filter at every lead time: −1.84 % at 1 month ($p < 10^{-6}$) and −0.80 to −0.38 % at 2 to 6 months ($p < 0.01$). Per basin at a 1-month lead time, after false-discovery-rate control, the split helps no basin and hurts none of the 234 tested; the loss is diffuse. The first fold’s training window ends in May 2019 and holds fewer than 12 GRACE-FO months, so all 234 of its basin-fold fits fall back to the single variance (Sect. 3.4). Every scored month is a GRACE-FO month, and a variance fit to the GRACE-FO months alone, fewer than seven years of them in any fold, is estimated less well than one fit to the whole record. We keep the single-variance filter.

4.3 Forecast models built on the filter

Table 3 scores the own-basin models against the filter. KF-R1, which adds one regression coefficient on the current filtered state, is weaker than KF at 1 and 2 months and stronger at 3 to 6 months, by +0.44 to +2.66 % ($p < 10^{-12}$); both read the same filtered state, and neither uses information that persistence does not have. KF-R12, which reads twelve months of the filtered state instead of one, improves on KF at every lead time, by +0.63 to +1.81 % ($p < 0.02$). KF-R12E adds the twelve-month forcing window and gains +7.65 % over KF at 1 month (95 % interval +4.7 to +10.1 %), +5.39 to +2.09 % at 2 to 5 months ($p < 0.001$) and a non-significant +0.87 % at 6 months ($p = 0.12$). Relative to damped persistence this is +12.2 % at 1 month and +13.7 % at 2 months (Table 2), the largest margins in the study. KF-R12E is the best own-basin model in standardized units at 1 to 4 months; at 5 and 6 months KF-R1, with one coefficient on the current filtered state, is better by less than 1 % (Appendix B). The forcing window improves on KF-R1 through 4 months and not beyond.

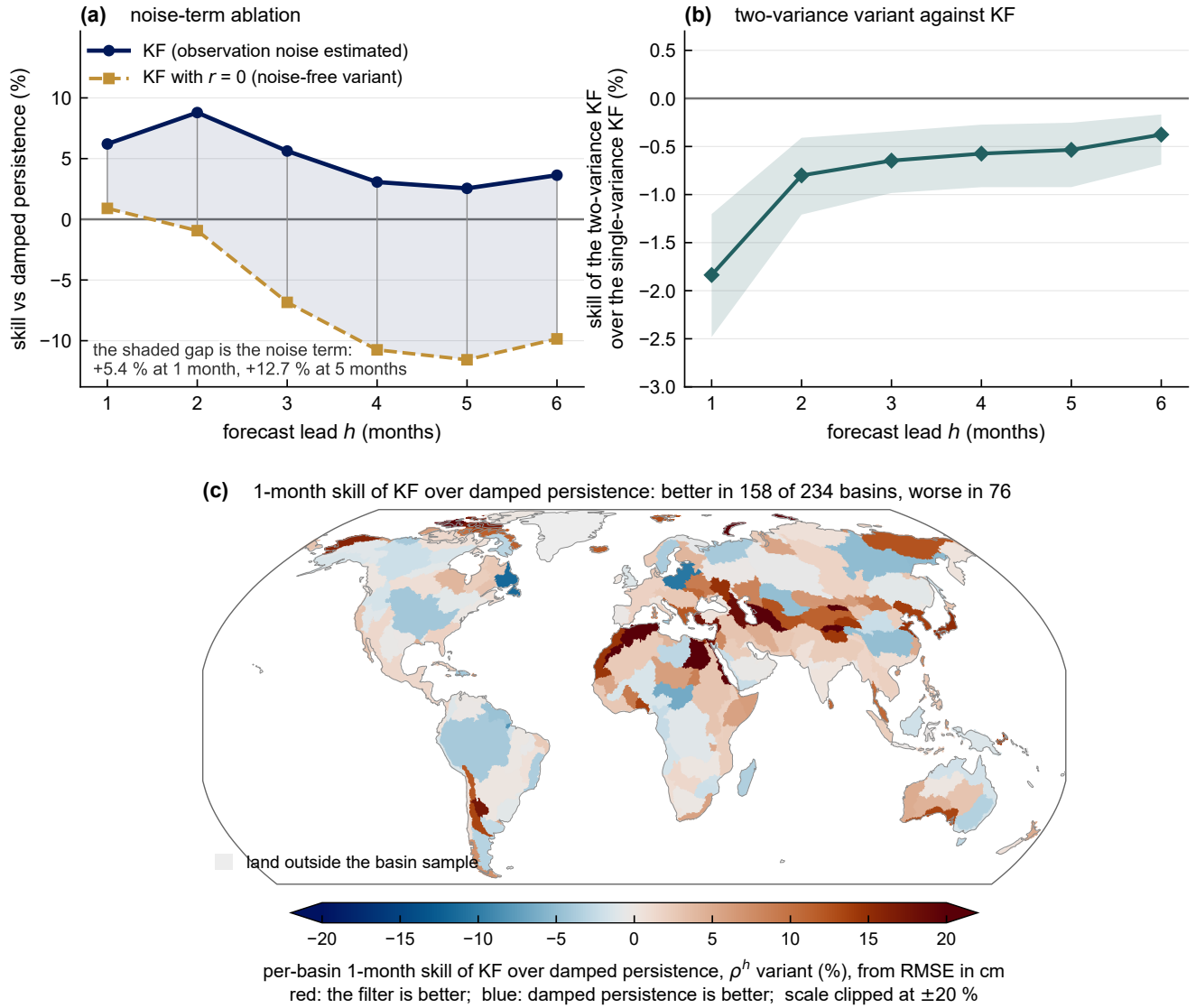


Figure 4. Attribution of the filter’s margin. (a) Skill (%) against the regression damped-persistence variant at every lead time for KF and for the noise-free variant (KF with $r = 0$). The shading is the gap between the two curves, which is the contribution of the noise term, and carries no confidence interval. Because the reference here is the regression variant at every lead time, KF reads +6.21 % at a 1-month lead time rather than the +5.0 % of Table 2, which uses the stronger ρ^h variant at that lead time. (b) The two-variance variant against the single-variance KF by lead time, with moving-block bootstrap 95 % confidence intervals; negative means the split hurts. (c) Per-basin 1-month skill of KF over the ρ^h damped-persistence variant on the 234 basins, computed from RMSE in centimeters; KF is better in 158 basins and damped persistence in 76. The color scale is clipped at ± 20 %, and land outside the basin sample is pale grey.

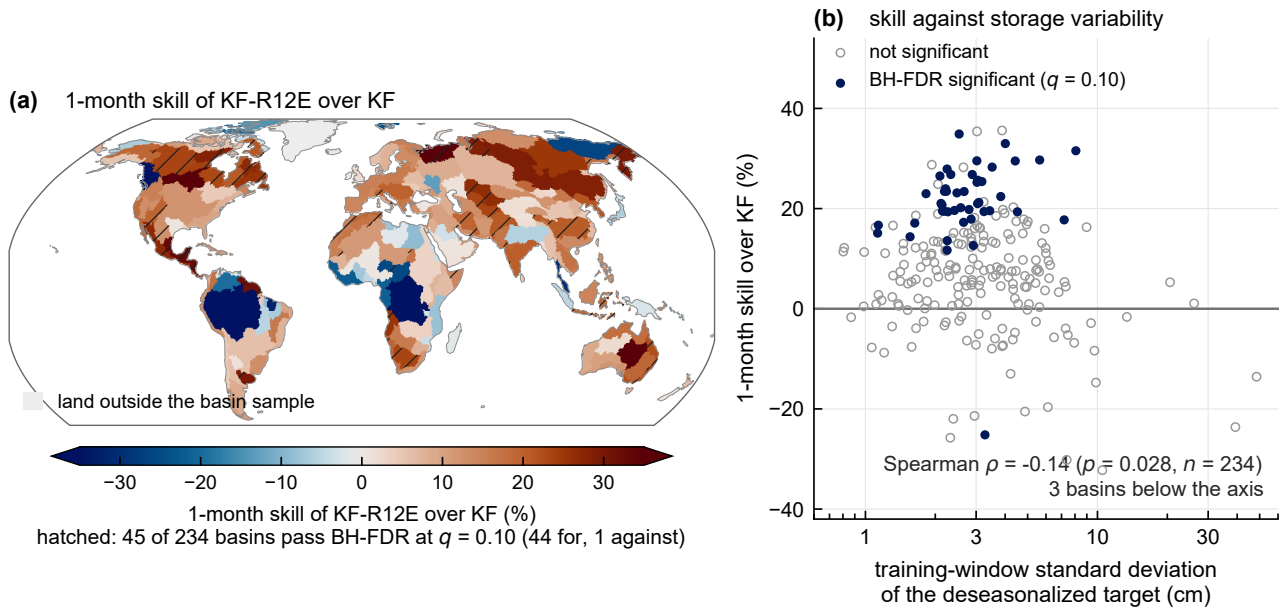


Figure 5. Where the forcing window helps. (a) Per-basin 1-month skill (%) of KF-R12E over KF on the 234 basins; basins significant under Benjamini–Hochberg false-discovery-rate control at $q = 0.10$ are hatched, and land outside the basin sample is pale grey. (b) The same skill against each basin’s training-window storage standard deviation in centimeters of equivalent water height; filled markers are the 45 significant basins (44 helped, 1 hurt), and three basins fall below the lower axis limit of -42% and are counted on the panel. The Spearman correlation is -0.14 . The color scale in (a) is clipped at $\pm 35\%$.

The gain from the forcing window is not uniform across basins. Figure 5 maps the per-basin 1-month skill of KF-R12E over KF and plots it against each basin’s training-window storage standard deviation. The relation is weak across the sample (Spearman $\rho = -0.14$), but the 45 basins that are significant under false-discovery-rate control, 44 helped and 1 hurt, all have training-window standard deviations below 9 cm, and none of the highest-variance basins gains significantly. One standardized unit is a small physical amount in a low-variance basin, so the forcing history explains a larger share of it. This property makes the ranking of the models sensitive to how basins are weighted (Sect. 4.5).

None of the nonlinear models improves on KF-R12E given the same twelve months of history (Table 4 and Fig. 6). At 1 to 3 months KF-R12E has the lowest standardized RMSE of every model and seed. The MLP, which sees the identical feature vector, is significantly worse in every seed at every lead time ($p < 10^{-5}$), and the residual MLP and both LSTM seeds fall between KF-R12E and KF-R1E. Because the LSTM holds out the last 15 % of its training months for early stopping, its comparison with a ridge fit on all training months is not matched on training data. We therefore refit KF-R12E on the LSTM’s training months. Against the two-seed LSTM ensemble the refit ridge is ahead by $+1.02\%$ at 1 month ($p = 0.094$), $+2.76\%$ at 2 months and $+2.27\%$ at 3 months (both $p < 10^{-8}$). It is indistinguishable from the ridge fit on all months (differences within

Table 4. Nonlinear models against KF-R12E: standardized pooled RMSE at lead times of 1 to 3 months on the same basin-months (n as in Sect. 3.2). Lower is better. Seeds are listed separately; the LSTM ensemble is scored in the text through the comparison matched on training months.

Forecast	$h=1$	$h=2$	$h=3$
KF	1.0490	1.1954	1.2734
KF-R1E	1.0254	1.1930	1.2685
KF-R12	1.0448	1.1865	1.2622
KF-R12E	1.0081	1.1627	1.2462
LSTM, seed 0 / seed 1	1.0132 / 1.0188	1.1758 / 1.1851	1.2578 / 1.2630
MLP, seed 0 / seed 1	1.0235 / 1.0291	1.1783 / 1.1935	1.2602 / 1.2799
Residual MLP, seeds 0 / 1 / 2	1.0148 / 1.0156 / 1.0145	1.1760 / 1.1788 / 1.1773	1.2635 / 1.2530 / 1.2540

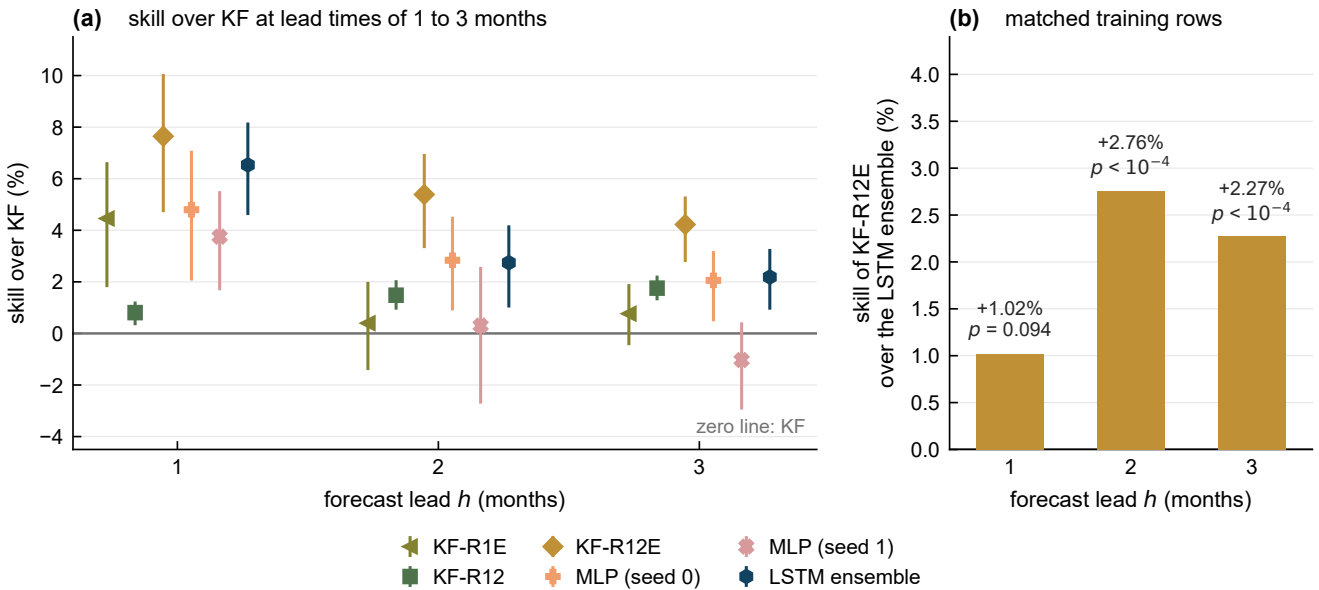


Figure 6. Nonlinear models against KF-R12E. (a) Skill (%) over KF at lead times of 1 to 3 months for KF-R1E, KF-R12, KF-R12E, the two seeds of the MLP and the two-seed LSTM ensemble, with moving-block bootstrap 95 % confidence intervals. The residual MLP appears in Table 4 only. (b) Skill of KF-R12E over the LSTM ensemble when the ridge is refit on the LSTM’s training months, with the Diebold–Mariano p value.

0.2%, $p \geq 0.19$). A linear model extracts the information in twelve months of forcing; under the training budget used here the nonlinear models extracted no more.

4.4 Comparison with GRACE-FCast on CSR and JPL

Table 5 and Fig. 7 compare our forecasts with GRACE-FCast on the fully covered basins. On CSR, over 209 basins and 60 months (12,540 basin-months per lead time), KF, with no forcing input, has lower error than the full product at a 1-month lead time, by 16.4% (95 % interval +4.9 to +26.6%). It ties at 2 months and has higher error from 3 months on, by 16.3 to 35.4% (290 $p < 0.002$). Against the non-seasonal component, which removes the seasonal and trend re-extrapolation, KF wins at 1 and 2 months (+28.4 and +7.5%), ties at 3 months and loses from 4 months on. KF-R12E, our best own-basin model at these lead times, follows the same win, tie and loss pattern as KF against both product variants, with larger 1-month margins (+21.5 and +32.8%) and interpolated zero crossings slightly later (Fig. 7). On JPL, over 67 basins and 59 months (3,953 basin-months per lead time), only KF is compared with GRACE-FCast; the ridge models were not run on that product. The pattern is the (295 CSR one shifted one lead time later. Against the full product KF wins at 1 month, ties at 2 and 3 months and loses from 4 months on; against the non-seasonal component it wins at 1 and 2 months, ties at 3 and 4 months and loses from 5 months on. Its 1-month margins there are large, +43.7% against the full product and +52.9% against the non-seasonal component, in which case KF's mean squared error is less than half the product's.

Basin by basin, the 1-month result is a near-even split that tips toward GRACE-FCast as the lead time grows. On CSR the full (300 product has the lower error in 98 of the 209 basins against KF at 1 month and in 157 at 6 months, and in 87 against KF-R12E at 1 month and 162 at 6 months. The pooled 1-month margin therefore reflects the size of the gain in the basins where we win rather than its breadth.

The 1-month result has a structural explanation. GRACE-FCast takes no lagged storage as input, so at a 1-month lead time, where the most recent observation is the strongest available predictor, every forecast that starts from that observation has (305 information the product lacks. We call such forecasts the persistence class; it includes damped persistence, KF and every model built on KF. Damped persistence itself beats the full product at 1 month, by +11.9% on CSR and +43.2% on JPL, and loses to it from 2 months on CSR and from 3 months on JPL (Table 5). The filter's own contribution at 1 month is its margin over damped persistence: +5.0% on CSR (Table 2) and a tie on JPL (+0.6%, $p = 0.60$; Sect. 4.1). The 1-month margin over the product is much larger on JPL than on CSR. The fully covered JPL set consists of 67 large basins with no small ones, and the (310 JPL member of GRACE-FCast was trained on an earlier release than the one we score it against (Sect. 2.4); we do not separate the two.

4.5 Sensitivity to basin weighting

Every ranking above is in standardized units, where each basin counts equally. In pooled raw centimeters, where the largest basins dominate, the five forecasts reported in that unit differ by about half a centimeter at most, and the differences among (315 the filter-based models are untested (Fig. 2b; Appendix C). KF-R12 has the lowest value at 1 to 3 months and KF-R1 at 4 to 6 months, and KF-R12E, the best model in standardized units, is 0.1 cm behind the filter at 1 month. Damped persistence has the highest RMSE at every lead time in this weighting as well. The filter's margin over persistence therefore does not depend on the weighting, but the forcing window's gain does: it lives in the low-variance basins of Fig. 5, which the standardized metric

Table 5. Comparison with GRACE-FCast on the fully covered basins. Skill (%) of our forecast over GRACE-FCast at each lead time h (months), upper value, with the Diebold–Mariano p value below it; positive means our forecast has the lower error. CSR: 209 basins, 60 months, 12,540 basin-months per lead time. JPL: 67 basins, 59 months, 3,953 basin-months per lead time, same convention. The last two rows count the CSR basins, of 209, in which the full product has the lower error at $h = 1$ and $h = 6$.

Pair	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
<i>CSR</i>						
KF vs. full	+16.4	−6.0	−16.3	−23.2	−28.9	−35.4
	0.0022	0.23	0.0014	3.9×10^{-5}	1.7×10^{-6}	1.7×10^{-8}
KF vs. non-seasonal	+28.4	+7.5	−4.0	−13.4	−20.3	−27.1
	2.7×10^{-10}	0.018	0.13	4.3×10^{-6}	7.8×10^{-9}	5.9×10^{-10}
KF-R12E vs. full	+21.5	−1.8	−12.1	−19.6	−26.4	−34.6
	4.0×10^{-4}	0.74	0.032	0.0017	7.3×10^{-5}	4.8×10^{-7}
KF-R12E vs. non-seasonal	+32.8	+11.2	−0.2	−10.1	−18.0	−26.4
	2.4×10^{-10}	0.0021	0.94	0.0032	6.7×10^{-6}	3.3×10^{-8}
Damped persistence vs. full	+11.9	−16.9	−30.9	−42.2	−47.9	−54.3
	0.029	0.0011	6.5×10^{-8}	2.7×10^{-10}	6.0×10^{-11}	2.0×10^{-12}
<i>JPL</i>						
KF vs. full	+43.7	+9.8	−6.2	−19.1	−32.1	−41.9
	7.1×10^{-14}	0.087	0.31	0.0017	1.2×10^{-6}	1.5×10^{-8}
KF vs. non-seasonal	+52.9	+22.2	+7.0	−6.2	−18.0	−27.9
	1.3×10^{-13}	2.1×10^{-5}	0.15	0.16	1.1×10^{-4}	2.8×10^{-6}
Damped persistence vs. full	+43.2	+4.1	−19.0	−39.2	−53.1	−65.3
	1.2×10^{-13}	0.49	0.0047	2.7×10^{-7}	1.1×10^{-9}	8.9×10^{-12}
Basins, full product beats KF	98					157
Basins, full product beats KF-R12E	87					162

JPL values are converted from the product-first skill s of the deposited tables as $1 - 1/(1 - s)$, the reciprocal of the mean-squared-error ratio.

counts in full and the centimeter metric barely counts. Any ranking of models on this target has to state which weighting it refers to; in this paper “best own-basin model” means lowest pooled standardized error.

4.6 Neighboring-basin predictors

Spatial and graph-based TWSA methods assume that one basin’s history helps forecast another’s future: Li and Kusche (2026) shift predictors across space, Arzoumanidis et al. (2026) build a proximity and correlation basin graph for reconstruction, and Steidl and Zhu (2025) forecast with a hierarchical basin graph. We tested the simplest version of that assumption: whether one other basin’s filtered state, propagated to the target month and added as a feature to KF-R1, improves the forecast. On CSR

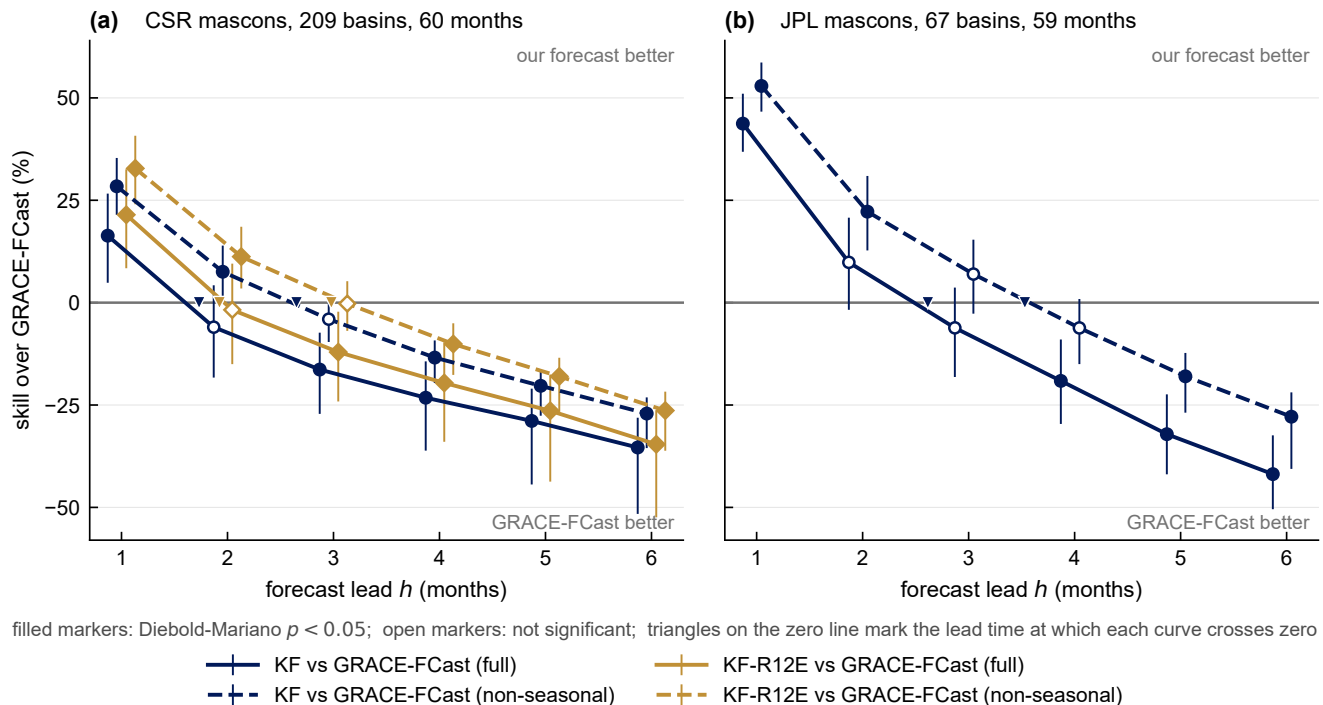


Figure 7. Comparison with GRACE-FCast on the fully covered basins. (a) CSR, 209 basins and 60 months: skill (%) of KF and of KF-R12E over GRACE-FCast, full and non-seasonal, by lead time, with moving-block bootstrap 95 % confidence intervals; positive means our forecast has the lower error. (b) JPL, 67 basins and 59 months, KF only. Filled markers are significant at $p < 0.05$, open markers are not, and the triangles on the zero line mark the interpolated lead time at which each curve crosses zero. Damped persistence, not drawn, also has lower error than the full product at a 1-month lead time on both products (Table 5); the 1-month result is shared by every forecast that starts from the last observation.

a correlation-selected neighbor was worth +0.31% over KF-R1 at 1 month, ahead of all 50 seed-matched random neighbor graphs and all 99 amplitude-adjusted Fourier-transform surrogates (Schreiber and Schmitz, 1996), and nothing at 2 to 6 months. On JPL none of this replicates: every neighbor variant is worse than KF-R1 at every lead time, the correlation-selected neighbor beats none of the 50 random graphs at any lead time, the distance-selected neighbor beats at most 26 of 50, and the real graph
330 beats none of the 99 surrogates. Because the effect reverses on a second processing of the same observations, we draw no conclusion from it.

5 Discussion

5.1 What the benchmark means for reported skill

The contribution of this paper is a benchmark, not a forecasting method. A persistence benchmark is meant to represent no
335 information beyond memory, but for a noisy observable it leaves part of that memory unused: on deseasonalized GRACE
basin series the unused part is 5.0 to 8.8 % at the lead times where most data-driven studies report their skill. A study scoring
against damped persistence concedes this margin, one scoring against raw persistence a further 16 to 23 %, and one scoring
against climatology far more (Table 2). The forecasting studies cited in Sect. 1 use a climatology, often implicit in a Nash–
Sutcliffe score, or another forecast system. Skill reported that way on observed basin TWSA therefore includes a margin that
340 a three-parameter filter would also have achieved, and cannot on its own be read as information beyond the storage record.

In physical units the margin is modest. At a 1-month lead time the pooled RMSE falls from 5.32 to 5.13 cm of equivalent
water height and the per-basin median from 2.74 to 2.66 cm (Appendix C). The gain is largest where the anomaly is small and
the observation noisy (Fig. 4c), which are the arid and small basins in which GRACE-based drought indicators are least certain
(Save et al., 2016); in large humid basins such as the Amazon the filter and damped persistence are nearly the same forecast
345 (Fig. 3d). The benchmark therefore changes the accounting of learned skill more than it changes the error of the basins that
dominate centimeter metrics.

The comparison with GRACE-FCast shows what the change of benchmark does in practice. GRACE-FCast is a capable
system: at lead times of 4 to 6 months every model we built has 10 to 35 % higher mean squared error, because its forcing
history carries information that no own-basin model has. At a 1-month lead time, however, it loses to every forecast tested here
350 that starts from the observation it does not use, damped persistence included, and at 2 months its non-seasonal component loses
to the filter. Neither statement is visible in a full-signal correlation score. Li and Kusche (2026) document the same sensitivity
to the target from the other side: they report correlation above 0.8 over about 72 % of global land at a 1-month lead time for the
full signal but over only 20 % once the seasonal cycle and trend are removed, and attribute the difference to the extrapolated
components. Most of the correlation in a full-signal GRACE forecast score is carried by terms that are extrapolated rather than
355 predicted, and Appendix C shows the same for our own forecasts. Forecast systems for this variable can be compared on the
deseasonalized target against a filter benchmark; full-signal correlation does not discriminate between them.

Our result complements that of Nie et al. (2025) (Sect. 1): their comparison, with lagged storage excluded, asks how well
forcing predicts storage; ours, with lagged storage included, finds that the comparison is decided by how the storage is filtered,
and that once it is filtered a linear model of the forcing history matched every nonlinear model we trained (Sect. 4.3). Both
360 results make linear models the model to beat on this target; ours adds that the benchmark itself should be a filter. The argument
applies to observed TWSA: targets that are themselves model output, reconstructions or gap fills, such as the HydroGlobe
target of Nie et al. (2025) or the products of Humphrey and Gudmundsson (2019), Yin et al. (2023) and Mo et al. (2022),
contain no satellite observation noise to filter, so neither the overstatement of skill over persistence nor the filter benchmark
applies to them.

365 5.2 Initial conditions versus forcing

Seasonal hydrological forecasting settled a structurally identical question with physically based models: skill at short lead times comes from initial conditions and skill at long lead times from meteorological forcing, with a crossing between them (Wood and Lettenmaier, 2008; Shukla and Lettenmaier, 2011; Koster et al., 2010; Yossef et al., 2013; Pechlivanidis et al., 2020). Those results were established for streamflow and soil moisture in ensemble streamflow prediction experiments. The comparison of Sect. 4.4 finds the same shape on satellite-observed storage between two real forecast systems, one built entirely on the observed state and one built entirely on forcing. On both mascon solutions the filter wins at short lead times and the forcing-driven product from 3 to 5 months on (Table 5); the short-lead half of that shape belongs to the persistence class rather than to the filter, since damped persistence wins at 1 month as well. We read the crossing as consistent with the initial-conditions-versus-forcing partition, not as a decomposition of skill into sources: the two systems differ in architecture, training data, gap handling and predictor sets at once, so what is measured is each system’s lead-time profile.

Below the crossing, three parameters filter the initial condition and none of the architectures we trained improved on a linear model of the filtered state. Above it, the twelve-month ERA5-Land window carried own-basin skill through 4 months (Sect. 4.3) and GRACE-FCast’s spatially shifted forcing history carried it through the 6 months we scored. A forecast that hands over from the filter to a forcing-driven model near the crossing would have to choose that lead time on data not used in this study.

5.3 Limitations

None of our models uses forecast forcing; the ERA5-Land window ends at the issue month. GRACE-FCast’s advantage from 3 or 4 months on CSR and from 4 or 5 months on JPL is the advantage of a forcing history read by a model trained on it, and the benchmark we propose has nothing to offer at those lead times beyond what the filtered state carries.

385 The 1-month advantage is also not available operationally. GRACE-FO solutions are released two to three months after the observation month (Mo et al., 2025), so a forecast issued from the latest available observation corresponds to a real-time lead time of 3 to 4 months, which on CSR is at or beyond the lead time at which GRACE-FCast overtakes the filter. Our comparison is between two hindcasts on a common issue-date convention; it does not reproduce the situation of a user who needs next month’s storage today.

390 The JPL comparison covers the filter only; the ridge models were not run on that product, so the two-product claim applies to the benchmark alone. The fully covered JPL set consists of 67 large basins, and the JPL member of GRACE-FCast was trained on an earlier release than we score it against, so the size of the 1-month margin there should not be compared with CSR’s. Relatedly, the filter’s 1-month margin over damped persistence does not replicate on JPL: the filter ties damped persistence there at 1 month and reproduces the margin at 2 to 6 months (Sect. 4.1). Our reading, that JPL’s processing leaves less month-to-month noise for the filter to remove, is consistent with the smaller persistence error on that product but is not tested directly.

All results are for basin averages. Most published data-driven TWSA forecasts are gridded (Li et al., 2020, 2024; Li and Kusche, 2026), where the noise fraction per cell is larger than in a basin average. The filter’s margin over persistence may be larger at grid scale, but we have not tested it, and (ρ, q, r) would then have to be estimated per cell.

The filter’s definition across the 2017–2018 gap follows from its prediction step, which runs whenever an update is skipped; no scored forecast is issued across the gap (Sect. 3.2), so this property is not tested here. Similarly, the noise term is not identified as instrument noise: 259 of 1170 basin-fold fits push r to its lower bound (Sect. 3.4), and the AR(1) plus white-noise decomposition absorbs unmodeled dynamics into r . The ablation attributes the margin to filtering within the model class; what is filtered out is a mixture of instrument noise and fast hydrology whose shares we do not estimate.

The ranking of models above the filter depends on basin weighting. The forcing window’s gain is concentrated in low-variance basins that pooled centimeter error barely counts (Sect. 4.5), while the filter’s margin over persistence holds in both weightings. The comparison with GRACE-FCast is not symmetric. Their training uses the gap reconstruction of Yin et al. (2023), our 1-month win is one that damped persistence shares and partly reflects a benchmark fit to exactly this target definition, and the compared sample excludes fold 5 and the basins their grid does not cover.

Finally, the evidence rests on two mascon products of one set of observations, five test folds all in the post-2019 climate, and per-basin estimates based on 83 months at a 1-month lead time. The LSTM, MLP and residual MLP were trained with the fixed architectures of Sect. 3.5, early stopping and no hyperparameter search, with two seeds each for the first two and three for the third; seed spread at 1 month is comparable to the gaps in Table 4, which is why no single seed is reported as a result. Under that budget they did not beat the linear model; whether a tuned sequence model would be is untested.

6 Conclusions

For basin-average, deseasonalized GRACE water storage, persistence of the last observation is a weaker benchmark than the storage record alone can support. A per-basin Kalman filter with an AR(1) state and additive observation noise, fit with three parameters, beats the stronger damped-persistence variant at every lead time on the CSR product and at 2 to 6 months on the JPL product, and the margin is the noise term: the same model refit without it has almost no margin. We propose the filter as the benchmark forecast for this task, and we recommend that data-driven forecasts of observed GRACE storage report skill against it rather than against persistence.

Three consequences follow. Reported skill over persistence or climatology on this target includes a margin that the filter would also achieve and should be re-read with that margin removed. Above the filter, a linear model of twelve months of filtered state and forcing is the model to beat at short lead times, and its gain depends on whether basins are weighted equally or by variance. And the lead-time profile against GRACE-FCast places the value of the storage record at 1 to 2 months and the value of forcing history from 3 to 5 months on; a forecast that hands over from one to the other near that lead time is the obvious next system to test.

Code and data availability. CSR GRACE and GRACE-FO mascon solutions are available from the Center for Space Research (<https://www2.csr.utexas.edu/grace/>) and JPL mascon solutions from NASA PO.DAAC (<https://podaac.jpl.nasa.gov/>). ERA5-Land data are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu/>) (Muñoz-Sabater et al., 2021). Climate indices are available from NOAA PSL (<https://psl.noaa.gov/data/climateindices/>). The GRACE-FCast hindcast is available from PANGAEA (Li, 2025). The analysis code is at <https://github.com/ishaankejriwal/graceseespaper>. The benchmark forecasts, the per-prediction files and every result table used in this paper, together with a manifest mapping each table and figure to its source file, will be deposited on Zenodo before submission at <https://doi.org/10.5281/zenodo.XXXXXXX>.

Appendix A: Model names and repository identifiers

The result files in the data repository identify each model by the identifier used in the analysis code. Table A1 maps the names of Table 1 to those identifiers, which appear nowhere else in the paper. Seed suffixes (`_s0`, `_s1`, `_s2`) denote single training seeds and `_ens` the seed ensemble.

Appendix B: The full benchmark tables

Table B1 gives, for every forecast in Table 2, the pooled standardized RMSE and the Diebold–Mariano p value against the stronger damped-persistence variant at all six lead times on the CSR product. Table B2 gives the corresponding skills and p values on the JPL product, where the ridge models built on the filter beyond KF-R1 were not run.

Appendix C: Conventional metrics in centimeters

Table C1 reports the five forecasts of Sect. 4.5 in the metrics the literature uses: pooled RMSE and per-basin median RMSE in centimeters of equivalent water height, and per-basin median correlation and Nash–Sutcliffe efficiency, on the deseasonalized anomaly and on the full signal. The full-signal forecast adds each fold’s frozen train-window climatology to the predicted anomaly. Because forecast and observation share that climatology, the centimeter errors are identical in the two spaces; only correlation and efficiency change, and the change is the share of a full-signal score carried by the extrapolated seasonal cycle and trend. At a 1-month lead time, damped persistence alone reaches a median full-signal correlation of 0.86 and a median full-signal efficiency of 0.72, values that would read as strong skill in a full-signal evaluation.

Author contributions. I.K. designed the study, wrote the analysis code, performed the experiments, and wrote the manuscript.

Competing interests. The author declares no conflict of interest.

Table A1. Model names used in the paper and the corresponding identifiers in the deposited result files.

Paper name	Identifier(s)
Climatology	climatology_zero
Persistence	persistence
Damped persistence, ρ^h and regression variants	damped_persistence_rho, damped_persistence_reg
Pooled lag ridge, and index variant	ridge_own_lags, ridge_own_plus_indices
Per-basin lag ridge	ridge_own_perbasin
KF	kalman_ar1
KF, $r = 0$	kalman_r0
KF, two variances	kalman_mission
KF-R1	kalman_own_ridge (ridge_own in the sequence-model files)
KF-R12	ridge_own_flat12
KF-R12E	ridge_own_era5_flat12; refit on the LSTM's training months ridge_own_era5_flat12_train85
KF-R1E	ridge_own_era5
LSTM	lstm_own_era5_s0, lstm_own_era5_s1, lstm_own_era5_ens
MLP	mlp_own_era5_flat12_s0, mlp_own_era5_flat12_s1
Residual MLP	resmlp_own_era5_s0, resmlp_own_era5_s1, resmlp_own_era5_s2
Neighbor ridge, correlation- and distance-selected	kalman_corr_top1, kalman_geo_top1
GRACE-FCast, full and non-seasonal	li_lstm_full, li_lstm_nonseas
Fully covered basin set	joint_full_cells; matched set all_matched

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external funding.

Table B1. CSR product: pooled standardized RMSE (upper value) and Diebold–Mariano p value against the stronger damped-persistence variant (lower value) for every forecast of Table 2 at lead times h of 1 to 6 months, on the basin-months of Sect. 3.2. The reference row has no test.

Forecast	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
Climatology	1.6163	1.6354	1.6525	1.6698	1.6881	1.7055
	6.0×10^{-39}	1.8×10^{-41}	3.3×10^{-36}	1.2×10^{-34}	2.8×10^{-35}	1.7×10^{-34}
Persistence	1.1825	1.3863	1.4442	1.4837	1.5224	1.5877
	1.4×10^{-11}	4.7×10^{-11}	3.7×10^{-11}	1.7×10^{-7}	2.7×10^{-4}	5.9×10^{-4}
Damped persistence, weaker variant	1.0831	1.2552	1.3519	1.4354	1.4953	1.5478
	2.1×10^{-4}	0.13	2.6×10^{-19}	1.3×10^{-23}	2.3×10^{-21}	7.9×10^{-22}
Damped persistence, stronger variant	1.0761	1.2516	1.3108	1.3573	1.4132	1.4754
	–	–	–	–	–	–
Pooled lag ridge	1.0698	1.2214	1.2770	1.3324	1.3906	1.4570
	0.15	2.4×10^{-5}	3.7×10^{-6}	4.3×10^{-5}	0.0026	0.013
Pooled lag ridge, index variant	1.0708	1.2230	1.2766	1.3316	1.3906	1.4578
	0.24	7.0×10^{-5}	3.0×10^{-6}	3.2×10^{-5}	0.0026	0.015
Per-basin lag ridge	1.0596	1.2139	1.2826	1.3433	1.4105	1.4776
	1.6×10^{-5}	2.2×10^{-12}	1.8×10^{-7}	8.5×10^{-5}	0.48	0.70
KF	1.0490	1.1954	1.2734	1.3363	1.3951	1.4484
	3.5×10^{-12}	6.3×10^{-19}	5.7×10^{-19}	2.6×10^{-11}	7.9×10^{-7}	2.0×10^{-9}
KF-R1	1.0499	1.2014	1.2706	1.3204	1.3764	1.4369
	1.5×10^{-11}	5.7×10^{-19}	1.7×10^{-19}	4.9×10^{-18}	6.7×10^{-12}	1.0×10^{-13}
KF-R12	1.0448	1.1865	1.2622	1.3241	1.3846	1.4438
	5.8×10^{-13}	1.9×10^{-20}	1.6×10^{-19}	1.5×10^{-16}	1.3×10^{-10}	2.7×10^{-11}
KF-R12E	1.0081	1.1627	1.2462	1.3148	1.3804	1.4421
	3.3×10^{-18}	2.1×10^{-20}	4.0×10^{-17}	3.4×10^{-15}	2.5×10^{-9}	2.5×10^{-12}

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Table B2. JPL product: pooled skill (%) against the stronger damped-persistence variant (upper value) and Diebold–Mariano p value (lower value) at lead times h of 1 to 6 months, 228 basins, five folds ($n = 19,152$ basin-months at $h = 1$). The reference is the ρ^h variant at $h = 1$ and 2 and the regression variant at $h = 3$ to 6. The reference row has no test.

Forecast	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$	$h=6$
Climatology	−374.3 6.3×10^{-40}	−184.4 3.3×10^{-40}	−141.3 6.2×10^{-36}	−119.3 3.6×10^{-35}	−95.0 1.8×10^{-35}	−75.5 8.5×10^{-36}
Persistence	−4.4 0.021	−7.1 0.0042	−4.7 0.050	−3.3 0.21	−2.1 0.45	+0.4 0.88
Damped persistence, weaker variant	−0.9 2.1×10^{-13}	−1.3 1.2×10^{-7}	−6.9 3.2×10^{-9}	−15.0 4.2×10^{-15}	−14.0 2.9×10^{-15}	−12.2 2.3×10^{-13}
Damped persistence, stronger variant	0.0 −	0.0 −	0.0 −	0.0 −	0.0 −	0.0 −
Pooled lag ridge	+0.5 0.62	+1.7 0.18	+2.2 0.058	−0.5 0.57	−1.1 0.18	−0.9 0.31
Pooled lag ridge, index variant	+0.7 0.47	+2.0 0.17	+2.8 0.042	+0.1 0.90	−0.5 0.63	−0.6 0.62
Per-basin lag ridge	+1.2 0.33	+3.2 0.0059	+2.3 0.0087	−0.6 0.43	−1.6 0.081	−2.1 0.058
KF	+0.6 0.60	+6.3 2.7×10^{-10}	+4.6 2.9×10^{-9}	+1.5 0.017	+2.4 9.8×10^{-5}	+4.7 8.9×10^{-11}
KF-R1	+0.3 0.80	+5.2 4.7×10^{-9}	+5.1 2.5×10^{-10}	+3.3 4.1×10^{-6}	+3.5 9.4×10^{-9}	+4.2 8.3×10^{-14}

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Table C1. Conventional metrics for the five forecasts reported in centimeters: pooled RMSE and per-basin median RMSE in centimeters of equivalent water height, and per-basin median correlation (CC) and Nash–Sutcliffe efficiency (NSE) on the anomaly and on the full signal, over 234 basins, at lead times h of 1 to 6 months. The damped-persistence rows use the stronger variant at each lead time.

Forecast	h	Pooled RMSE (cm)	Median RMSE (cm)	CC anom.	CC full	NSE anom.	NSE full
Damped persistence	1	5.321	2.74	0.62	0.86	0.33	0.72
	2	6.156	3.32	0.46	0.81	0.09	0.59
	3	6.383	3.57	0.39	0.79	0.02	0.54
	4	6.447	3.65	0.34	0.78	−0.01	0.51
	5	6.689	3.78	0.30	0.77	−0.04	0.48
	6	6.935	3.96	0.23	0.75	−0.08	0.43
KF	1	5.128	2.66	0.64	0.86	0.36	0.74
	2	5.650	3.23	0.50	0.82	0.18	0.63
	3	5.949	3.46	0.43	0.80	0.06	0.56
	4	6.161	3.57	0.37	0.78	0.00	0.52
	5	6.413	3.72	0.30	0.76	−0.02	0.48
	6	6.690	3.84	0.24	0.74	−0.05	0.44
KF-R1	1	5.133	2.66	0.64	0.86	0.36	0.74
	2	5.691	3.26	0.50	0.82	0.17	0.63
	3	5.931	3.46	0.43	0.80	0.06	0.56
	4	6.064	3.55	0.37	0.78	0.01	0.53
	5	6.288	3.68	0.31	0.76	−0.01	0.49
	6	6.603	3.82	0.25	0.74	−0.05	0.45
KF-R12	1	5.118	2.66	0.64	0.86	0.36	0.74
	2	5.634	3.19	0.51	0.82	0.18	0.63
	3	5.915	3.39	0.45	0.81	0.08	0.57
	4	6.107	3.52	0.37	0.78	0.03	0.53
	5	6.366	3.72	0.30	0.76	−0.02	0.49
	6	6.678	3.85	0.26	0.74	−0.05	0.44
KF-R12E	1	5.231	2.56	0.69	0.88	0.42	0.76
	2	5.741	3.12	0.55	0.83	0.22	0.65
	3	6.002	3.37	0.48	0.81	0.10	0.58
	4	6.163	3.48	0.39	0.78	0.04	0.54
	5	6.463	3.69	0.33	0.76	−0.02	0.48
	6	6.774	3.87	0.27	0.74	−0.05	0.43